
Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

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Abstract

1 Quantizing a large language model reduces serving cost and costs some accu-
2 racy. Determining how much normally requires re-running a full benchmark suite,
3 which is the expense quantization was meant to avoid. We ask whether the accu-
4 racy delta can be predicted instead. Using 817 published evaluations spanning 38
5 base checkpoints, 8 model families and 6 quantization schemes, we report three
6 results. First, a *negative* one: per-model prediction carries almost no signal. Un-
7 der leave-one-family-out validation the best predictor, a per-scheme mean, beats
8 a global mean by 0.03pp of MAE (95% checkpoint-bootstrap CI [0.01, 0.05]),
9 and both ridge regression and gradient boosting are *worse* than guessing the av-
10 erage, because evaluation noise accounts for at least a third of the variance on 8
11 of 11 benchmarks. Second, a positive one: conditioning only on the quantization
12 scheme yields per-scheme conformal-style intervals with 90.1% empirical cover-
13 age (118/131, 95% CI [83.6, 94.6]) on checkpoints never used in construction.
14 Those checkpoints cover 4 of the 6 schemes, and W4A16 supplies 80 of the 131
15 rows. Each (scheme, size-band) cell returns one of three verdicts: trusted; insuf-
16 ficient evidence, where the interval is still shown but the scheme is demoted and
17 flagged; or refused, where the interval is withheld. Currently 9 of 17 cells are in-
18 sufficient evidence and 0 are refused. Third, and most consequential for practice:
19 the published corpus is outcome-truncated. Every configuration in it is one that
20 someone chose to release. We rented an A100 and measured deliberately-faulted
21 configurations, observing losses to -39.5pp against a published worst case of
22 -8.86pp —measured, we stress, only on models of 1.5B and below, and on config-
23 urations engineered to fail rather than typical ones. This is an existence proof that
24 far worse outcomes exist than the corpus contains; it does not estimate how com-
25 mon they are. We argue the resulting artifact should be described as a calibrated
26 historical baseline rather than a predictor, and we state precisely why its coverage
27 guarantee does not transfer to a practitioner’s own untuned recipe. Code, data and
28 the provenance of every reported figure are provided as supplementary material.

29 1 Introduction

30 Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8
31 wants one number: how much accuracy will this cost? In practice the only way to know is to run the
32 evaluation.

33 This paper asks whether that number can be forecast from the public record. Model publishers
34 release quantized checkpoints with benchmark tables comparing the compressed model against its
35 dense parent. Aggregated, these are hundreds of paired measurements of exactly the quantity of
36 interest (850 extracted in our harvest).

37 Our contributions:

- 38 1. **A negative result** (§5). Per-model prediction of the accuracy delta does not work, and
39 we show why: the target is dominated by evaluation noise. We report this as the primary
40 finding rather than burying it.
- 41 2. **A calibrated envelope that does work** (§6), conditioned on scheme alone, with measured
42 coverage on held-out families and a three-state verdict for each (scheme, size-band) cell:
43 *trusted*; *insufficient evidence*, where the interval is still shown but the scheme is demoted
44 and the cell flagged as one the tool cannot judge; or *refused*, where the interval is withheld.
45 Currently 9 of 17 cells are insufficient evidence and 0 are refused.
- 46 3. **An existence proof that the corpus omits severe outcomes** (§7), via GPU experiments on
47 deliberately-faulted configurations—data the published record structurally cannot contain.
48 It shows that losses far beyond anything published are reachable on small models. It does
49 not estimate how large the bias is: our attempt at a correction (in the supplement) did not
50 work well enough to ship.
- 51 4. **An audit** (§9) recording errors we found in our own work, including ones that changed
52 published numbers and claims.

53 2 Related Work

54 **Predicting evaluations.** BenchPress Zeng and Papailiopoulos [2026] establishes the mechanism
55 closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and
56 decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks
57 is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is
58 a model’s absolute score; ours is the paired delta between a model and its compressed self, and
59 quantized checkpoints do not appear in their paper.

60 **Conformal prediction on compressed models.** Tong et al. [2026] apply conformal prediction
61 directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The estimand
62 differs: they evaluate predictive uncertainty (coverage and set size) *after* running the compressed
63 model. Ours is constructed over historical accuracy deltas to give an interval *before* running any-
64 thing. Their Figure 2 plots the same quantity we predict, $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$, but measures it.
65 Their finding that compression decouples accuracy from uncertainty bounds what an accuracy-only
66 estimate like ours can claim (§8).

67 **Quantization methods.** The checkpoints we harvest are produced with GPTQ Frantar et al. [2022]
68 and SmoothQuant Xiao et al. [2022] via llm-compressor vLLM Project [2024]. Kumar et al. [2024]
69 fit a functional form that predicts degradation from training and inference in varied precisions, using
70 pretraining runs of models up to 1.7B parameters. We fit no such law: our envelope is empirical, per
71 scheme, over published post-training-quantized checkpoints.

72 **The source of our data.** Kurtić et al. [2024] report over 500,000 evaluations of quantized Llama-
73 3.1 models, finding recovery above 99% of the average OpenLLM v1 score for every scheme tested.
74 As far as we can tell, that work does not address whether the published set of quantized models is a
75 biased sample. Characterising that gap is the contribution of §7.

76 **Selection bias.** Survivorship bias has been identified and measured in a widely used retrieval
77 benchmark Gupta and MacAvaney [2022]. Conformal prediction’s guarantee is known to fail un-
78 der selection: conditional on a selection event, calibration data are no longer exchangeable with a
79 test point Barber et al. [2023], Jin and Ren [2024]. We apply this directly to our own artifact in §8.

80 3 Data

81 We harvest published model cards into rows of (model, scheme, benchmark, $\text{acc}_{\text{before}}$, $\text{acc}_{\text{after}}$), yield-
82 ing 850 rows, of which 817 remain after dropping 33 whose baseline accuracy is below 20% (near
83 chance, where a delta is uninformative), across 38 base checkpoints, 8 families and 6 schemes.

84 **An integrity gate.** Cards typically print a *Recovery %* column alongside the before/after pair.
 85 This is redundant information, and we exploit it: for every three-column row we check recovery $\approx$
 86 $100 \times$ after/before to a tolerance derived from the card’s own printed precision, and reject rows that
 87 disagree. This caught spurious values that our parser produced under column-order variation be-
 88 tween card templates, and internally inconsistent cards. Rejections are recorded rather than silently
 89 dropped.

90 **A structural limitation, stated at the outset.** All rows come from a single publisher using a
 91 single toolchain. Holding out a model family tests transfer across *models*; nothing in this design
 92 tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

93 4 Method

94 **Conformal-style intervals.** We use the residual-quantile construction of split conformal re-
 95 gression Lei et al. [2018]: given residuals $\{s_i\}_{i=1}^n$ and level α , the interval half-width is the
 96 $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, which under a proper calibration/test split gives exact finite-
 97 sample coverage under exchangeability. Where n is too small for that index to exist, we return
 98 ∞ rather than a misleadingly finite band. The shipped per-scheme interval fits its centre and its
 99 half-width on the *same* rows (an in-sample residual-quantile band), so the split-conformal guaran-
 100 tee holds only for the explicit calibration-set variant used inside the leave-one-family-out audit; the
 101 empirical prospective figure below is what carries the shipped intervals.

102 **Mondrian (group-conditional) calibration.** Calibrating separately within categories is the Mon-
 103 drian construction of Vovk et al. [2005], applied to regression by Boström and Johansson [2020]. A
 104 single marginal interval advertised at 90% covers only 62% of NVFP4 rows under leave-one-family-
 105 out (recomputed live)—the users who most need it. We therefore calibrate per scheme.

106 **A three-state verdict.** For each (scheme, size-band) cell we measure coverage directly, and the
 107 tool returns one of three verdicts. A cell is *trusted* when measured coverage is adequate and ad-
 108 equately supported. It is *refused*—INSUFFICIENT CALIBRATION, no interval—when coverage is
 109 measurably poor: the checkpoint-bootstrap interval on its coverage lies wholly below the 85% line,
 110 on at least 3 checkpoints. It is *insufficient evidence* when the cell rests on fewer than 3 distinct
 111 checkpoints, or when a checkpoint-level cluster bootstrap 90% interval on its coverage straddles the
 112 85% line. In that case the evidence cannot support either of the other two verdicts. The tool still
 113 prints the all-sizes interval for such a cell, but demotes the scheme to the lowest tier and states that
 114 the cell cannot be judged; only a refused cell has its interval withheld.

115 Coverage is scored *one-sided*: how often the true delta stayed at or above the interval’s lower bound.
 116 The risk this tool bounds is accuracy loss, and a model that beats its envelope has exposed no one
 117 to anything, so counting that as a failure would flag cells that are merely outperforming. Comparing
 118 a one-sided measurement against the printed 90% is the *lenient* reading, not a flattering one: the
 119 interval is symmetric, so the permitted 10% is split across two tails, and pooled one-sided coverage,
 120 across all 817 rows with each row scored under 7 different calibration families, is 95.1%.

121 Currently 0 of 17 cells are refused. Two cells were flagged under the earlier two-state rule, and nei-
 122 ther is a confident refusal under the three-state one. `w4a16|<2B` had one-sided coverage of 70.1%
 123 over 11 rows (two-sided 68.8%), but those rows come from only 2 checkpoints. `w8a16|>10B` had
 124 81.1% over 31 rows from 5 checkpoints (two-sided 73.7%), and its bootstrap interval, [47, 100]%,
 125 straddles the threshold. Both are now *insufficient evidence*. We had scoped the change to those
 126 two cells; applying the rule to every cell moved 9 of 17 into that state, including several whose
 127 point-estimate coverage looked comfortable. The same measurement and the same rule judge each
 128 scheme’s all-sizes coverage: 1 of the 6 schemes is insufficient evidence at that level and 0 is refused.
 129 The one-sided scoring and the three-state rule were proposed by an external commenter who identi-
 130 fies as an AI account collaborating with the OpenLanguageModel maintainers; we recomputed each
 131 of their figures from the raw rows before adopting either change, and one of their descriptions did
 132 not survive that check.

133 **A support floor.** Coverage measured on many benchmarks from one checkpoint is not independent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct
 134 checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.
 135

136 5 The negative result

137 Under leave-one-family-out validation, row-weighted MAE:

	predictor	MAE (pp)	beats global mean?
	per-scheme mean (shipped)	0.7227	yes
	global mean (baseline)	0.7545	—
138	per-(scheme $\times$ benchmark)	0.7607	no
	ridge regression	0.7639	no
	per-benchmark mean	0.7647	no
	gradient boosting	0.7878	no

139 The per-scheme mean’s advantage is small: 0.03pp, with a checkpoint-bootstrap 95% CI of
 140 [0.01, 0.05]pp, and it is positive in 7 of the 8 held-out families. It is a distinguishable but modest gain.
 141 Ridge and gradient boosting are both *worse* than predicting the global average. Adding model size,
 142 bit-width, method, benchmark identity and headroom each degraded out-of-family performance.

143 **Why.** Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta
 144 should be approximately zero—gives a noise floor that accounts for anywhere from 15% of the
 145 observed variance to all of it across 11 benchmarks. On GSM8K the deltas of supposedly lossless
 146 schemes have a standard deviation of 2.1pp, while the quantity being predicted has mean -0.39 pp.
 147 The target is a difference of two noisy quantities that largely cancel: the numerically smallest and
 148 hardest-to-predict summary of quantization damage available. Two cautions on the noise figures.
 149 They assume near-lossless schemes have a true delta near zero, so they are upper bounds on noise.
 150 And 5 of the 11 estimates rest on 10 or fewer rows; one, at 124%, exceeds the total variance, which
 151 we attribute to sampling error rather than treat as a finding.

152 **Implication.** The entire fitted artifact is 18 numbers. A raw per-scheme historical quantile band,
 153 with no model and no conformal machinery, is a serious competitor: it looked better covered in-
 154 sample, though under the strict protocol that advantage disappears (§9). What conformal contributes
 155 is a finite-sample guarantee and a principled rule for small- n schemes, not accuracy. We therefore
 156 describe the result as a *calibrated historical baseline*, not a predictor.

157 6 Results

	scheme	90% interval (pp)	worst observed	% losing >3 pp	evals
	fp8	[-1.11, +0.48]	-1.90pp	0.0%	80
	w8a8_int	[-1.71, +1.07]	-3.32pp	1.0%	194
158	fp8_dynamic	[-1.44, +1.34]	-8.72pp	1.0%	200
	w8a16	[-0.98, +0.81]	-3.04pp	1.2%	86
	w4a16	[-2.87, +1.40]	-8.86pp	5.7%	193
	nvfp4	[-4.10, +1.87]	-8.12pp	15.6%	64

159 **Prospective validation.** After freezing the method we harvested checkpoints not used during de-
 160 velopment. Under a strict protocol that excludes any family present in training and any card whose
 161 format informed a parser fix, coverage is $118/131 = 90.1\%$, 95% Clopper–Pearson CI [83.6, 94.6],
 162 against a nominal 90%. A from-scratch, standard-library-only reimplementation sharing no code
 163 with the pipeline reproduces this figure with zero disagreements. That checks the computation, not
 164 the validity of the design.

165 **What the prospective set does and does not cover.** The 131 strict rows come from 19 quantized
 166 checkpoints, not 131 independent observations: a bootstrap that resamples checkpoints gives a 95%

interval of $[83.2, 94.7]\%$, close to the row-level one. The rows cover only 4 of the 6 schemes; FP8 and NVFP4 have no prospective rows, so the figure says nothing about them.

	scheme	strict rows	checkpoints	covered (%)
	w4a16	80	11	87.5
	w8a8_int	36	5	91.7
	fp8_dynamic	11	2	100.0
	w8a16	4	1	100.0

W4A16, the most common scheme, dominates the pooled figure and sits below the nominal 90%; the other schemes rest on few checkpoints. Over all 186 harvested prospective rows, before the strict filter, coverage is 90.3%, so the filter did not manufacture the headline.

What the envelope cannot do. 0 of 6 intervals exclude zero: the tool can never state that a scheme *will* cost accuracy. And 19 of 25 observed losses worse than 3pp fell *below* their interval floor—when real damage occurs, the interval usually misses it. This is the single most important caveat in the paper.

7 The measured left tail

The corpus contains only configurations someone chose to publish. To measure what it structurally cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

Result. Worst measured loss -39.5pp , against -8.86pp as the worst loss anywhere in the published corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor instead of per-group.

Scope, stated plainly. This tail was measured only on models of 1.5B and below. It establishes that catastrophic damage is reachable; it is not a measured bound for larger models.

A confound we found and did not resolve. Re-running MMLU under the published protocol (5-shot, letter-scored) recovers accuracies in the range published on the cards: our Qwen2.5-0.5B-Instruct scores 47.20 against the Instruct card’s 46.83 (the base-model card gives 47.57), and our Qwen2.5-1.5B-Instruct scores 59.57. The harness question closed. Our control run then compared Instruct-model checkpoints against Red Hat’s *base-model* W4A16 publications, because no Instruct W4A16 checkpoints are published: our correctly-configured Instruct control is $2.6\times$ more damaging than the base-model published W4A16 (mean gap -3.78pp , $z = 2.9$ and 5.3). This is not like-for-like—Instruct models could plausibly be more fragile under 4-bit quantization than their base counterparts—so the ratio is indicative rather than a clean estimate. A matched Instruct-vs-Instruct control is deferred future work. The pattern we observed is consistent with approximately $171\times$ less calibration data than production defaults, though the attribution is not independently tested. Consequently, excess-over-control measures *bad recipe versus mediocre recipe*, not versus production. It is an indicative anchor, not a bound, and we decline to publish a corrected point estimate: the correction moves by 5.70pp depending on the anchor set, and we have only 6 rows from two models.

8 Limitations

The guarantee is conditional, and absent where it matters most. Split conformal gives exact coverage under exchangeability between calibration and test. Our calibration set is the set of configurations a publisher chose to release—a selection event. Conditional on selection, calibration points are not exchangeable with an arbitrary test point and the guarantee does not transfer Barber et al. [2023], Jin and Ren [2024]. It holds for a new checkpoint drawn from the same publication process. It carries **no guarantee** for a recipe the reader tuned themselves and that no publisher would have released—which is the case a practitioner most wants covered.

208 **Accuracy is not the whole of deployment risk.** Tong et al. [2026] show compression frequently
 209 decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets inflate. A
 210 scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We measure none
 211 of that.

212 **One publisher, one toolchain, and why.** See §3. Our own control arm (§7) is direct evidence
 213 that practitioner and calibration volume can dominate scheme choice. We measured how confining
 214 this is: surveying 10 publishers and 260 quantized model cards, 10 of 222 cards from the other 9
 215 publishers carry paired before/after evaluations, 9 of them Intel’s, so others do publish them. What
 216 the other publishers almost never provide is a *Recovery* column, which is what lets each row be
 217 checked against $100 \times$ after/before and rejected on disagreement: of their rows, 6 pass that gate,
 218 all from 1 card, against 147 from the control publisher. The corpus is single-publisher not because
 219 others publish nothing, but because the control publisher’s cards are, by a wide margin, the ones
 220 published in a self-verifying form.

221 **What can and cannot be re-run.** The supplement recomputes every tagged figure from the data
 222 files it ships. It cannot re-run the faulted-configuration experiments (they need a GPU; the recorded
 223 results are included), the harvest from raw model cards (the cards are not redistributed), or the
 224 prospective-card checks (those files are not included), and re-running the low-rank comparison men-
 225 tioned in the supplement needs a checkout of BenchPress’s released code.

226 **Rows are not independent.** Up to sixteen benchmark rows come from a single quantization run.
 227 We mitigate with group-conditional calibration and the support floor, and note that a bootstrap over
 228 the 19 quantized checkpoints gives [83.2, 94.7]%, close to the row-level interval, but the unit of anal-
 229 ysis remains a row. The leave-family-out coverage measurement has the same dependence: each
 230 row is scored once under each of 7 calibration families, so the 5719 evaluations are not indepen-
 231 dent, and every coverage judgment in this paper resamples whole checkpoints rather than counting
 232 evaluations.

233 9 Audit

234 We record errors found in our own work, including those that changed published claims. This
 235 section exists because the alternative—reporting only what survived—would misrepresent how the
 236 result was obtained.

237 Errors that changed a published number.

- 238 • *A retracted headline.* An early claim that held-out models came from families never used
 239 in construction was wrong; they were new checkpoints of families already present. The
 240 strict figure is 118/131.
- 241 • *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was
 242 matched as Math-Lvl-5, contaminating 7 test rows with scores from a different bench-
 243 mark.
- 244 • *A centring mismatch.* Calibration measured residuals about the stratum mean while in-
 245 tervals centred on the scheme mean, mis-sizing every calibrated width. Found only by
 246 independent reimplementations.
- 247 • *A column-order fault.* One card template orders columns [recovery, base, quant], which our
 248 parser turned into a spurious -68pp delta. The integrity gate rejected 10 of 11 rows on that
 249 card, passing only the one where base = quant made the check degenerate.
- 250 • *A retracted single-publisher claim.* We had asserted that one publisher was close to the
 251 only source of paired evaluations. A census of 10 publishers found 10 paired-evaluation
 252 cards outside the control publisher, 9 of them Intel’s; what differs is whose rows can be
 253 verified (§8).
- 254 • *A bootstrap on the wrong statistic.* The checkpoint-level bootstrap in our coverage audit re-
 255 sampled the two-sided indicator while the verdict was judged on the one-sided statistic. An
 256 external commenter asked for it to wrap the one-sided figure, which exposed the mismatch
 257 (§4).

- 258 • *Inflated pair counts.* An earlier version of the tool reported the number of (test row, cali-
259 bration family) evaluations as if it were the number of rows: with 8 families each row is
260 scored 7 times, so 5719 evaluations came from only 817 rows. Every row-count claim was
261 recomputed on `distinct_rows`.
 - 262 • *Two levels judged differently.* Cells were judged on the one-sided coverage statistic and
263 refused when the point estimate fell below the threshold, but schemes had been judged on
264 the two-sided statistic and refused only when the entire bootstrap interval fell below it. A
265 single classifier now judges both levels the same way, and the envelope’s refused list is a
266 computed copy of the ranking tool’s, checked by a test.
- 267 **A regression we nearly shipped.** An audit indicated that a raw asymmetric empirical band cov-
268 ered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization
269 damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch.
270 On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-
271 sided empirical index requires more calibration points than are often available. An infinite interval
272 covers by construction. On rows where both are finite, the original wins on coverage *and* width. The
273 change was reversed.
- 274 **An overclaimed guarantee.** The artifact was described as carrying a coverage guarantee without
275 qualification. Checking the method against the conformal literature rather than against our own
276 earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected
277 in §8.
- 278 **Provenance discipline.** Two parser fixes were informed by inspecting rows that were already in
279 the test set. Rather than silently retain them, we record which fixes were test-informed and exclude
280 the affected family from the strict headline.
- 281 **Mechanical verification.** The figures this paper reports as results are macros generated from a
282 claims registry that recomputes each value from source artifacts, and the accompanying code fails
283 a test if an outward-facing document contains an untraced numeric literal. The exceptions are his-
284 torical figures in the error list above (row counts and the -68pp delta), which are quoted from the
285 provenance record rather than recomputed.

286 10 Conclusion

287 Per-model prediction of quantization accuracy damage does not work on published data, and the
288 reason is largely measurement noise rather than model capacity. What does work is narrower: a
289 per-scheme calibrated envelope, validated at 90.1% coverage on unseen checkpoints for 4 of the 6
290 schemes, and that says which of its own cells it cannot judge. Each cell is trusted; or insufficient
291 evidence, in which case the interval is still shown but flagged and the scheme demoted; or refused,
292 in which case the interval is withheld. Today 9 of 17 cells are insufficient evidence and 0 are refused.
293 The more useful contribution may be the limitation itself—the published record of quantized models
294 is outcome-truncated, and we measured, on deliberately faulted recipes for models up to 1.5B, losses
295 more than four times worse than anything it contains. A risk estimate built on that record is a floor
296 on risk, not a ceiling, and should say so.

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